

Course Review: Group Theory and Galois Theory

MAT205 Abstract Algebra II

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Part I: Group Theory

Group Actions: Orbit and Stabilizer

Definition. An *action* of G on a set X is a map $G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x$, with $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. Equivalently, a homomorphism $G \rightarrow \text{Sym}(X)$.

Orbit and stabilizer.

$$\text{Orb}(x) = \{g \cdot x : g \in G\}, \quad \text{Stab}(x) = \{g \in G : g \cdot x = x\} \leq G.$$

Orbit-Stabilizer Theorem.

$$|\text{Orb}(x)| = [G : \text{Stab}(x)].$$

In particular, orbit sizes divide $|G|$; distinct orbits partition X .

Conjugacy Classes and the Class Equation

Definition. The *conjugation action* of G on itself is $g \cdot x = gxg^{-1}$. Its orbits are the *conjugacy classes*:

$$\text{cl}(x) = \{gxg^{-1} : g \in G\}.$$

The stabilizer of x is the *centralizer* $Z_G(x) = \{g \in G : gx = xg\}$.

Orbit-Stabilizer gives

$$|\text{cl}(x)| = [G : Z_G(x)] \text{ divides } |G|.$$

Class equation. Taking one representative x_i from each non-central class,

$$|G| = |Z(G)| + \sum_i [G : Z_G(x_i)],$$

where each $[G : Z_G(x_i)] > 1$. This is the main tool for p -group arguments (each term divisible by p forces $p \mid |Z(G)|$).

Cauchy's Theorem and the Sylow Theorems

Definition. A *Sylow p -subgroup* of G is a maximal p -subgroup. For $|G| = p^a m$ with $\gcd(p, m) = 1$: P is Sylow iff $|P| = p^a$. Write $n_p = |\text{Syl}_p(G)|$.

Cauchy's Theorem. If $p \mid |G|$, then G has an element of order p .

Sylow I. For each $1 \leq i \leq a$, G has a subgroup of order p^i . In particular, Sylow p -subgroups exist.

Sylow II. All Sylow p -subgroups of G are conjugate.

Sylow III. $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.

Corollary. $P \trianglelefteq G \iff n_p = 1$. Forcing $n_p = 1$ for some prime p shows G is not simple.

Hints: How to Force $n_p = 1$

Typical exam target: show G has a normal Sylow p -subgroup, i.e. $n_p = 1$. Standard methods:

(A) Sylow III alone. List divisors of m that are $\equiv 1 \pmod{p}$. If only 1 satisfies both, $n_p = 1$. *E.g.* $|G| = 15$: $n_5 \mid 3$, $n_5 \equiv 1 \pmod{5} \Rightarrow n_5 = 1$.

(B) Element counting. If Sylow p -subgroups have prime order, distinct ones intersect trivially and contribute $n_p(p - 1)$ elements of order p . Combine several such counts; if the total exceeds $|G|$, some n_p must equal 1. *Cue:* $|G|$ is a product of distinct primes.

(C) Small index / coset action. If $[G : P] = k$ is small, the action of G on G/P gives $G \rightarrow S_k$. When $|G| \nmid k!$, the kernel is nontrivial and forces P normal. *Cue:* $[G : P]$ smaller than the smallest prime dividing $|G|$.

(D) Residual p -subgroup. After counting elements of other orders, the leftover p^a elements are exactly one Sylow p -subgroup, hence unique.

p -Groups: Center and Order p^2

Theorem. If $|G| = p^k > 1$ (p prime), then $Z(G) \neq \{e\}$.

Corollaries.

- Every group of order p^2 is abelian.
- Every finite p -group is nilpotent.

Fundamental Theorem of Finite Abelian Groups

Every finite abelian group G has two canonical decompositions, both unique up to reordering of summands.

(1) Invariant factor form.

$$G \cong \mathbb{Z}/d_1 \oplus \mathbb{Z}/d_2 \oplus \cdots \oplus \mathbb{Z}/d_k, \quad d_1 \mid d_2 \mid \cdots \mid d_k.$$

The integers $d_1, \dots, d_k$ are the *invariant factors*; d_k is the exponent of G .

(2) Primary decomposition (elementary divisor form).

$$G \cong \bigoplus_{p \mid |G|} G_p, \quad G_p = \bigoplus_{i=1}^{r_p} \mathbb{Z}/p^{e_{p,i}}, \quad e_{p,1} \geq \cdots \geq e_{p,r_p} \geq 1.$$

Here G_p is the (unique, characteristic) Sylow p -subgroup of G . The prime powers $p^{e_{p,i}}$ are the *elementary divisors*.

Passage. G_p is the p -primary part of every invariant factor; conversely, invariant factors are assembled by aligning the primary partitions.

Worked Example: Invariant Factors vs. Elementary Divisors

Problem. Rewrite $G = \mathbb{Z}_{12} \oplus \mathbb{Z}_{18} \oplus \mathbb{Z}_{20}$ in both canonical forms.

Primary form. Split each summand by prime and collect:

$$G \cong \underbrace{\mathbb{Z}_4 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_2}_{G_2} \oplus \underbrace{\mathbb{Z}_9 \oplus \mathbb{Z}_3}_{G_3} \oplus \underbrace{\mathbb{Z}_5}_{G_5}.$$

Invariant factors. Align columns of primary parts (largest with largest), fill missing entries with 1, multiply down:

p	d_3	d_2	d_1
2	4	4	2
3	9	3	1
5	5	1	1
col	180	12	2

 $\implies G \cong \mathbb{Z}_2 \oplus \mathbb{Z}_{12} \oplus \mathbb{Z}_{180}.$

Worked Example: Classifying Abelian Groups of a Given Order

Problem. List all abelian groups of order 200, up to isomorphism.

Solution. $200 = 2^3 \cdot 5^2$.

- 2-part: one of $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $\mathbb{Z}_2^3$ (the 3 partitions of 3).
- 5-part: one of $\mathbb{Z}_{25}$, $\mathbb{Z}_5 \times \mathbb{Z}_5$ (the 2 partitions of 2).

The six groups are

$$\begin{aligned} &\mathbb{Z}_8 \times \mathbb{Z}_{25}, \quad \mathbb{Z}_8 \times \mathbb{Z}_5^2, \quad \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}, \\ &\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5^2, \quad \mathbb{Z}_2^3 \times \mathbb{Z}_{25}, \quad \mathbb{Z}_2^3 \times \mathbb{Z}_5^2. \end{aligned}$$

Composition Series and the Jordan–Hölder Theorem

Definition. A nontrivial group S is *simple* if its only normal subgroups are 1 and S .

Definition. A *composition series* is a subnormal series

$$G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_n = 1$$

with every factor G_i/G_{i+1} simple; n is the *composition length*. Every finite group has one.

Jordan–Hölder Theorem. Any two composition series of G have the same length, and after reordering, isomorphic factors: the multiset $\{G_i/G_{i+1}\}$ of composition factors is an invariant of G .

Worked Example: Composition Factors of an Abelian Group

Problem. Determine the composition factors of $A = \mathbb{Z}_{20} \times \mathbb{Z}_{45}$.

Solution. A is finite abelian, so its composition factors are cyclic of prime order, one for each prime factor of $|A|$ (with multiplicity). Since

$$|A| = 20 \cdot 45 = 900 = 2^2 \cdot 3^2 \cdot 5^2,$$

the composition factors are $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_3, \mathbb{Z}_5, \mathbb{Z}_5$.

Solvable and Nilpotent Groups

Solvable. G is *solvable* if it has a subnormal series

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = 1 \quad \text{with every } G_i/G_{i+1} \text{ abelian.}$$

Equivalent (finite G): every composition factor is cyclic of prime order; equivalently, no composition factor is non-abelian simple.

Nilpotent. G is *nilpotent* if its lower central series reaches 1:

$$G = \gamma_1(G) \supseteq \gamma_2(G) = [G, G] \supseteq \gamma_3(G) = [G, \gamma_2(G)] \supseteq \cdots = 1,$$

equivalently, its upper central series $1 = Z_0 \trianglelefteq Z_1 = Z(G) \trianglelefteq Z_2 \trianglelefteq \cdots$ reaches G .

Equivalent (finite G): G is the direct product of its Sylow subgroups (equivalently, every Sylow subgroup is normal).

Hierarchy. abelian $\Rightarrow$ nilpotent $\Rightarrow$ solvable. All strict: D_4 is nilpotent non-abelian; S_3 is solvable non-nilpotent.

A_n Is Simple for $n \geq 5$

Theorem. A_n is simple and non-abelian for every $n \geq 5$.

This fails below $n = 5$: $A_3 \cong \mathbb{Z}_3$ is abelian simple, and A_4 has a normal Klein-four subgroup $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$.

Corollary. S_n is solvable $\iff n \leq 4$. For $n \geq 5$, S_n has A_n as a non-abelian simple composition factor.

Worked Example: Proving a Non-Abelian Group Is Solvable

Problem. Show A_4 is solvable, and find its composition factors.

Setup. Inside A_4 (order 12), the *Klein four-group* is

$$V_4 = \{ e, (12)(34), (13)(24), (14)(23) \} \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

All three non-identity elements are the products of two disjoint 2-cycles in A_4 ; conjugation by any $\sigma \in A_4$ permutes them (as $\sigma(ab)(cd)\sigma^{-1} = (\sigma a \sigma b)(\sigma c \sigma d)$), so $V_4 \trianglelefteq A_4$.

Composition series.

$$A_4 \triangleright V_4 \triangleright \langle (12)(34) \rangle \triangleright 1,$$

with $\langle (12)(34) \rangle \trianglelefteq V_4$ since V_4 is abelian. Successive quotients:

$$A_4/V_4 \cong \mathbb{Z}_3, \quad V_4/\langle (12)(34) \rangle \cong \mathbb{Z}_2, \quad \langle (12)(34) \rangle \cong \mathbb{Z}_2.$$

All prime-order cyclic, so A_4 is solvable.

Characteristic Subgroups

Definition. $H \leq G$ is *characteristic* in G ($H \text{ char } G$) if $\alpha(H) = H$ for every $\alpha \in \text{Aut}(G)$.

Standard characteristic subgroups. $Z(G)$; $[G, G]$; each pA and $A[p]$ in a finite abelian group; each $p^k G$ in a finite abelian p -group.

Theorem. If $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.

In particular, every characteristic subgroup of G is normal (take $H = G$).

The Normalizer–Centralizer Theorem

Definitions.

$$N_G(H) = \{g \in G : gHg^{-1} = H\}, \quad Z_G(H) = \{g \in G : ghg^{-1} = h \ \forall h \in H\}.$$

Theorem. $Z_G(H) \trianglelefteq N_G(H)$, and

$$N_G(H)/Z_G(H) \hookrightarrow \text{Aut}(H).$$

Special case. For $H = \{x\}$: $N_G(\{x\}) = G$ and $Z_G(\{x\}) = Z_G(x)$, recovering conjugacy-class arithmetic via $[G : Z_G(x)] = |\text{cl}(x)|$.

Part II: Fields and Galois Theory

Irreducibility Toolbox for $\mathbb{Q}[x]$

Almost every Part-II problem hinges on irreducibility. Standard tests for $f \in \mathbb{Z}[x]$:

Eisenstein's criterion. If some prime p satisfies

$$p \mid a_0, a_1, \dots, a_{n-1}, \quad p \nmid a_n, \quad p^2 \nmid a_0,$$

then $f = a_n x^n + \dots + a_0$ is irreducible over $\mathbb{Q}$.

Rational Root Theorem. If $f \in \mathbb{Z}[x]$ has rational root p/q in lowest terms, then $p \mid a_0$ and $q \mid a_n$. Useful for degree 2 and 3: such f is irreducible over $\mathbb{Q}$ iff it has no rational root.

Reduction mod p . If $p \nmid a_n$ and $\bar{f} \in \mathbb{F}_p[x]$ is irreducible, then f is irreducible over $\mathbb{Q}$.

Gauss's Lemma. A primitive $f \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$ iff it is irreducible over $\mathbb{Z}$; the product of two primitive polynomials is primitive. (Bridges $\mathbb{Z}[x]$ - and $\mathbb{Q}[x]$ -irreducibility.)

Substitution. If $f(x+a)$ (some $a \in \mathbb{Z}$) is Eisenstein, so is f . Standard trick: $\Phi_p(x+1)$ is Eisenstein at p .

Simple Extensions and Kronecker's Construction

Simple extension. An extension of the form $F(\alpha)$; when α is algebraic with minimal polynomial m_α of degree n ,

$$F(\alpha) \cong F[x]/(m_\alpha), \quad F\text{-basis } \{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}.$$

Kronecker's Theorem. For any field F and any irreducible $f \in F[x]$, the quotient

$$K := F[x]/(f)$$

is a field, F embeds into K via constants, and the class $\bar{x} := x + (f) \in K$ is a root of f . So every irreducible polynomial has a root in some extension.

Consequences.

- Every $f \in F[x]$ has a splitting field: iterate Kronecker.
- Two roots of the same irreducible f over F give F -isomorphic simple extensions (isomorphism extension).

Tower Law: Degrees and Indices

Two multiplicative laws with the same shape, both used constantly.

Tower Law (fields). For $F \subseteq K \subseteq E$,

$$[E : F] = [E : K] \cdot [K : F].$$

Index formula (groups). For subgroups $K \leq H \leq G$,

$$[G : K] = [G : H] \cdot [H : K].$$

Special case ($K = 1$, G finite): $|G| = [G : H] \cdot |H|$ —Lagrange.

Worked example (degree). Compute $[\mathbb{Q}(\sqrt{6}, \sqrt[3]{10}) : \mathbb{Q}]$.

$[\mathbb{Q}(\sqrt{6}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt[3]{10}) : \mathbb{Q}] = 3$ (Eisenstein at $p = 2$ or $p = 5$). Let $d = [\mathbb{Q}(\sqrt{6}, \sqrt[3]{10}) : \mathbb{Q}]$. Tower Law through each subfield gives $2 \mid d$ and $3 \mid d$, so $6 \mid d$; multiplicativity across the joint generation gives $d \leq 6$. Hence $d = 6$.

Minimal Polynomial and Conjugates

Definition. For α algebraic over F , the *minimal polynomial* $m_\alpha \in F[x]$ is the monic generator of $\{f \in F[x] : f(\alpha) = 0\}$; it is irreducible, and $[F(\alpha) : F] = \deg m_\alpha$.

Definition. α, β are *conjugate over F* if $m_\alpha = m_\beta$; equivalently, they are roots of the same monic irreducible over F .

Isomorphism Extension. If α, β are conjugate over F , there is a unique F -isomorphism $F(\alpha) \rightarrow F(\beta)$ sending $\alpha \mapsto \beta$, and it extends to an automorphism of any normal closure containing both.

Consequence. The conjugates of α are exactly the roots of m_α in $\overline{F}$.

Worked Example: Listing Conjugates

Problem. List the conjugates of $\sqrt[5]{11}$ over $\mathbb{Q}$.

Solution. $x^5 - 11$ is irreducible over $\mathbb{Q}$ (Eisenstein, $p = 11$), so it is the minimal polynomial of $\sqrt[5]{11}$. Its five roots in $\mathbb{C}$ are

$$\zeta_5^k \sqrt[5]{11}, \quad k = 0, 1, 2, 3, 4, \quad \zeta_5 = e^{2\pi i/5}.$$

These are the conjugates of $\sqrt[5]{11}$ over $\mathbb{Q}$.

Algebraic Closure

Definition. A field $\overline{F}$ is an *algebraic closure* of F if $\overline{F}/F$ is algebraic and every nonconstant $f \in \overline{F}[x]$ splits over $\overline{F}$.

Theorem (Steinitz).

- Every field F has an algebraic closure.
- Any two algebraic closures of F are F -isomorphic.

(Proved via Zorn's lemma on the poset of algebraic extensions.)

Working inside a fixed $\overline{F}$ lets us speak unambiguously of “the” conjugates of α , “the” splitting field of f , and “the” normal closure of E/F .

Splitting Fields and Normal Extensions

Definition. E/F is a *splitting field* for $f \in F[x]$ if f splits over E and E is generated over F by the roots of f .

Definition. E/F (inside a fixed $\overline{F}$) is *normal* if every F -embedding $\sigma : E \rightarrow \overline{F}$ satisfies $\sigma(E) = E$.

Theorem (equivalences, E/F finite). TFAE:

1. every irreducible $p \in F[x]$ with a root in E splits in E ;
2. E is the splitting field over F of some $f \in F[x]$;
3. every F -embedding $E \rightarrow \overline{F}$ has image E .

Checking Non-Normality: a Standard Pattern

Criterion. If some irreducible $f \in F[x]$ has *one* root in E but does not *split* in E , then E/F is not normal.

Prototype. $\mathbb{Q}(\sqrt[3]{a})/\mathbb{Q}$ is not normal for any non-cube $a > 0$: it contains the real root $\sqrt[3]{a}$ of the irreducible $x^3 - a$, but not the non-real roots $\omega\sqrt[3]{a}$, $\omega^2\sqrt[3]{a}$.

Same pattern (real root of an irreducible with non-real conjugates) covers $\mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\sqrt[3]{3})$, $\mathbb{Q}(\sqrt[3]{5})$, ...

Separability and Galois Extensions

Definition. α is *separable* over F if m_α has no repeated roots in $\overline{F}$. E/F is separable if every $\alpha \in E$ is separable over F .

Automatic separability. In characteristic 0, and for any finite field.

Definition. E/F finite is *Galois* if it is normal and separable, equivalently

$$|\operatorname{Aut}(E/F)| = [E : F].$$

Its *Galois group* is $\operatorname{Gal}(E/F) := \operatorname{Aut}(E/F)$, of order $[E : F]$.

Finite Fields: Existence, Uniqueness, Multiplicative Group

Theorem. For each prime power $q = p^n$, there is a subfield $\mathbb{F}_q \subseteq \overline{\mathbb{F}_p}$ with exactly q elements; any two finite fields of order q are isomorphic.

Construction. $\mathbb{F}_{p^n} = \{a \in \overline{\mathbb{F}_p} : a^{p^n} = a\}$, the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$.

Theorem. Every finite subgroup of $F^\times$ (any field F) is cyclic. In particular $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

Frobenius and Subfields of $\mathbb{F}_{p^n}$

Frobenius. The map $\sigma(a) = a^p$ is an automorphism of $\mathbb{F}_{p^n}$, and

$$\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle \sigma \rangle \cong \mathbb{Z}_n.$$

Subfield criterion. $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n} \iff d \mid n$. Subfields are unique for each admissible order.

Root factorization.

$$x^{p^n} - x = \prod_{a \in \mathbb{F}_{p^n}} (x - a).$$

Consequence. $x^q - x + 1$ has no root in $\mathbb{F}_q$, so no finite field is algebraically closed.

Worked Example: Smallest Field Where $x^n - 1$ Splits

Problem. Find the smallest n such that $x^{15} - 1 \in \mathbb{F}_2[x]$ splits completely over $\mathbb{F}_{2^n}$.

Solution. $x^{15} - 1$ splits over $\mathbb{F}_{2^n}$ iff $\mathbb{F}_{2^n}^\times$ (cyclic of order $2^n - 1$) contains all 15th roots of unity, iff $15 \mid 2^n - 1$. Computing powers of 2 mod 15:

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8, \quad 2^4 = 16 \equiv 1 \pmod{15}.$$

The order of 2 mod 15 is 4, so the smallest such n is $n = 4$.

Cyclotomic Fields

Theorem. The minimal polynomial of $\zeta_n = e^{2\pi i/n}$ over $\mathbb{Q}$ is the cyclotomic polynomial Φ_n , irreducible of degree $\varphi(n)$. Hence

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n), \quad \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

via $\sigma_a : \zeta_n \mapsto \zeta_n^a$ for $a \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Structure. By CRT, $(\mathbb{Z}/n\mathbb{Z})^\times \cong \prod_p (\mathbb{Z}/p^{e_p}\mathbb{Z})^\times$, where each factor is cyclic for odd p^{e_p} and 2, 4, and $(\mathbb{Z}/2^k\mathbb{Z})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_{2^{k-2}}$ for $k \geq 3$.

Worked Example: Cyclotomic Galois Groups, Cyclic and Not

(a) Prime modulus. Is $\text{Gal}(\mathbb{Q}(\zeta_{11})/\mathbb{Q})$ cyclic?

Yes: $(\mathbb{Z}/11\mathbb{Z})^\times = \mathbb{F}_{11}^\times$ is the multiplicative group of the field $\mathbb{F}_{11}$, hence cyclic, of order 10.

(b) Composite modulus. Is $\text{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q})$ cyclic?

No: $\text{Gal}(\mathbb{Q}(\zeta_{12})/\mathbb{Q}) \cong (\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$, and every non-identity element squares to 1 mod 12:

$$5^2 = 25 \equiv 1, \quad 7^2 = 49 \equiv 1, \quad 11^2 = 121 \equiv 1 \pmod{12}.$$

So the group is $\mathbb{Z}_2 \times \mathbb{Z}_2$ — **not** cyclic.

Primitive Element Theorem

Theorem. Every finite separable extension E/F is simple: there exists $\gamma \in E$ with $E = F(\gamma)$.

In particular, every finite Galois extension is generated by a single element.

Practical form. For $E = F(\alpha_1, \dots, \alpha_n)$ separable over an infinite F , almost every $\mathbb{F}$ -linear combination $\gamma = c_1\alpha_1 + \dots + c_n\alpha_n$ works.

Example. $\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2} + \sqrt{3})$.

Fundamental Theorem of Galois Theory

Theorem. Let E/F be finite Galois, $G = \text{Gal}(E/F)$. The map $K \mapsto \text{Gal}(E/K)$ is a bijection from intermediate fields $F \subseteq K \subseteq E$ onto subgroups of G :

1. $K = E^{\text{Gal}(E/K)}$, and $\text{Gal}(E/E^H) = H$ for every $H \leq G$;
2. $[E : K] = |\text{Gal}(E/K)|$ and $[K : F] = [G : \text{Gal}(E/K)]$;
3. K/F is normal $\iff \text{Gal}(E/K) \trianglelefteq G$, and then $\text{Gal}(K/F) \cong G / \text{Gal}(E/K)$;
4. the correspondence **reverses inclusions**.

Worked Example: Galois Group of a Cubic Splitting Field

Problem. Let E be the splitting field of $x^3 - 3$ over $\mathbb{Q}$. Find $\text{Gal}(E/\mathbb{Q})$ with explicit generators.

Solution. $E = \mathbb{Q}(\sqrt[3]{3}, \omega)$, $\omega = e^{2\pi i/3}$. Tower Law:

$$[E : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{3}) : \mathbb{Q}] \cdot [\mathbb{Q}(\sqrt[3]{3}, \omega) : \mathbb{Q}(\sqrt[3]{3})] = 3 \cdot 2 = 6.$$

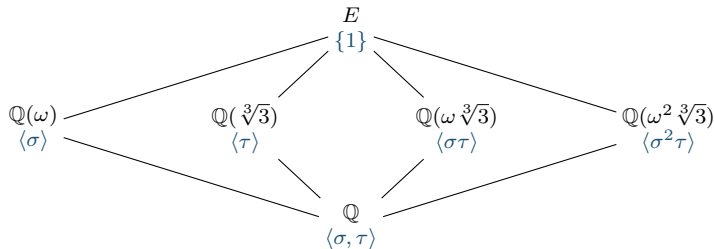
E is normal (splitting field) and separable (char. 0), so $E/\mathbb{Q}$ is Galois with $|\text{Gal}(E/\mathbb{Q})| = 6$. Let

$$\sigma : \sqrt[3]{3} \mapsto \omega \sqrt[3]{3}, \omega \mapsto \omega \text{ (order 3)}, \quad \tau : \sqrt[3]{3} \mapsto \sqrt[3]{3}, \omega \mapsto \omega^2 \text{ (order 2)}.$$

Then $\tau\sigma\tau^{-1} = \sigma^{-1}$, so $\langle \sigma, \tau \rangle \cong S_3$; since $|\langle \sigma, \tau \rangle| = 6$, $\text{Gal}(E/\mathbb{Q}) \cong S_3$.

Worked Example: Explicit Galois Correspondence

For $E = \mathbb{Q}(\sqrt[3]{3}, \omega)$ and σ, τ as before, the subgroup lattice of $\text{Gal}(E/\mathbb{Q}) \cong S_3$ and its fixed fields (field lattice drawn upside-down):



Only $\langle \sigma \rangle$ (order 3, index 2) is normal in S_3 , matching $\mathbb{Q}(\omega)/\mathbb{Q}$ being the unique normal proper subextension. The three order-2 subgroups are conjugate, not normal, matching the three non-normal cubic subfields.

Counting Irreducible Polynomials via Frobenius Orbits

Key idea. Monic irreducibles of degree d over $\mathbb{F}_q$ correspond to $\langle \text{Fr} \rangle$ -orbits of size d inside $\mathbb{F}_{q^d}$, where $\text{Fr}(\alpha) = \alpha^q$.

Degree d over $\mathbb{F}_q$, d prime. Every $\alpha \in \mathbb{F}_{q^d} \setminus \mathbb{F}_q$ has degree exactly d , so its orbit has size d . Hence

$$\#\{\text{monic irreducibles of degree } d\} = \frac{q^d - q}{d}.$$

Special case, $d = 2$: $(q^2 - q)/2$.

General formula (Gauss/Möbius).

$$\#\{\text{degree } n \text{ irreducibles over } \mathbb{F}_q\} = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d}.$$

Constructions by Straightedge and Compass

Setup. A point $z \in \mathbb{C}$ (identifying $\mathbb{R}^2$ with $\mathbb{C}$) is *constructible* from $\{0, 1\}$ if it can be obtained by finitely many straightedge-and-compass steps.

Theorem (Wantzel). z is constructible iff z lies in a tower

$$\mathbb{Q} = K_0 \subset K_1 \subset \cdots \subset K_r, \quad [K_{i+1} : K_i] = 2.$$

In particular $[\mathbb{Q}(z) : \mathbb{Q}]$ is a power of 2.

Classical impossibilities.

- Doubling the cube: $\sqrt[3]{2}$ has degree 3, not a power of 2.
- Trisecting a general angle: $\cos 20^\circ$ has degree 3.
- Squaring the circle: π is transcendental.

Gauss–Wantzel. The regular n -gon is constructible iff $n = 2^k p_1 \cdots p_s$ with distinct Fermat primes p_i .

Solvability by Radicals

Definition. A polynomial $f \in F[x]$ is *solvable by radicals* over F if its splitting field lies in a tower $F = K_0 \subset K_1 \subset \cdots \subset K_r$ with each $K_{i+1} = K_i(\sqrt[n_i]{a_i})$ for some $a_i \in K_i$.

Galois Criterion. Over a field of characteristic 0, f is solvable by radicals $\iff \text{Gal}(E/F)$ is a solvable group, where E is the splitting field of f over F .

The generic n -th degree equation. Its Galois group over the field of coefficients is S_n . For $n \geq 5$, S_n is not solvable (A_n is a non-abelian simple composition factor).
Hence:

Abel–Ruffini. No general formula in radicals solves the quintic (or any higher-degree generic polynomial).